

Worked Examples On Angles Of Elevation And Depression

QUESTIONS

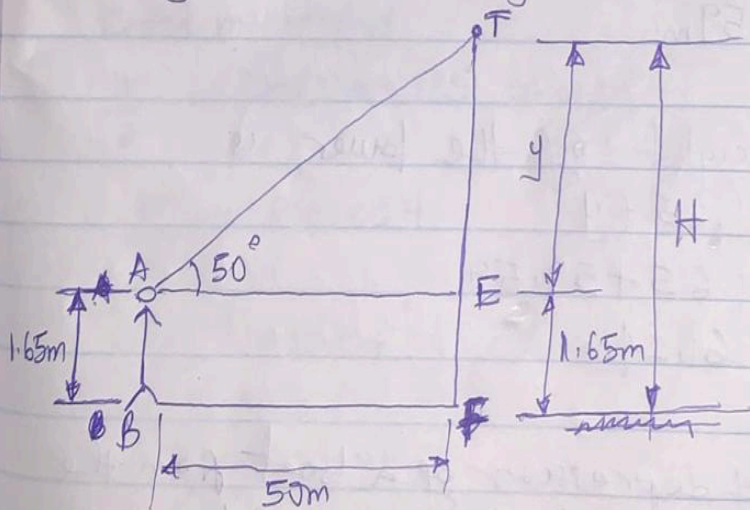
1) A woman 1.65m tall stood 50m away from the foot of a tower and observed that the angle of elevation of the top of the tower to be 50° . What is the height of the tower?

Solution

The things to consider while solving this question are:

- (i) height of the woman
- (ii) distance between the woman and the building
- (iii) angle of elevation of the top of the building
- (iv) height of the building

Let us consider these things and put them together in a diagram as shown below



where

AB = height of the woman

AE = normal line of sight

AT = elevated line of sight

T = Top of the tower

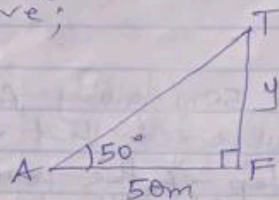
F = Foot of the tower

From the diagram;

$$AB = EF = 1.65\text{m}$$

$$TF = \text{height of the tower} = H$$

Let us focus on triangle ATE from above, we have;



$$\text{opp} = y, \theta = 50^\circ, \text{Adj} = 50\text{m}$$

Hence,

$$\tan \theta = \frac{\text{opp}}{\text{Adj}}$$

$$\tan 50 = \frac{y}{50}$$

Cross multiply

$$y = 50 \times \tan 50$$

$$y = 50 \times 1.1918$$

$$y = 59.59\text{m}$$

~~y~~

Hence, the height of the tower is

$$H = 1.65 + y$$

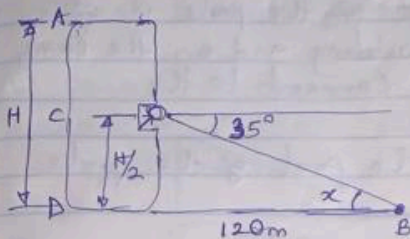
$$H = 1.65 + 59.59$$

$$H = 61.24\text{m}$$

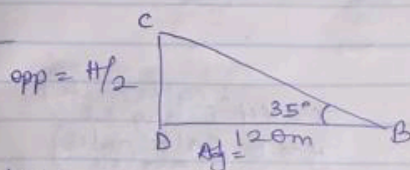
2) The angle of depression of a boat from the midpoint of a vertical cliff is 35° . If the

boat is 120m from the foot of the cliff,
Calculate the height of the cliff.

Solution



$x = 35^\circ$ (alternate angles)



where

$$CD = \frac{1}{2} \text{ of } AD$$

AD = height of the cliff = H

B = boat's position

To find the height of the cliff, we would use TOA.

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\tan 35 = \frac{H/2}{120}$$

Cross multiply

$$\frac{H}{2} = 120 \times \tan 35 = 120 \times 0.7002$$

$$\frac{H}{2} = 84.024$$

cross multiply

$$H = 2 \times 84.024$$

$$H = 168.048$$

$H \approx 168\text{m}$ (nearest whole number)

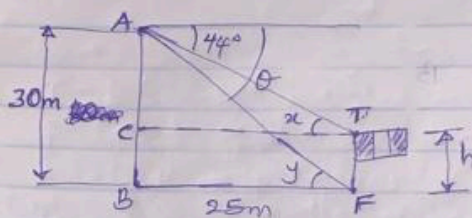
Therefore, the height of the cliff is 168m.

3) A woman looking out from the window of a building at a height of 30m, observed that the angle of depression of the top of a flag pole was 44° . If the foot of the pole is 25m from the foot of the building and on the same horizontal ground, find, correct to the nearest whole number, the

a) angle of depression of the foot of the pole from the woman

b) height of the pole

solution



$$x = 44^\circ \text{ (alternate angles)}$$

$$y = \theta \text{ (alternate angles)}$$

where

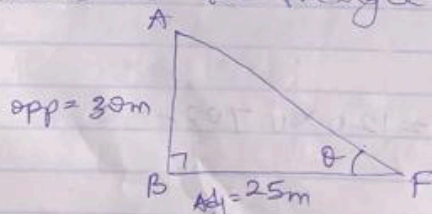
AB = height of the window above the ground

CB = height of the flagpole = h = FT

θ = angle of depression of the foot of the flagpole.

~~the height of the flag~~

a) Consider the triangle ABF



$$\tan \theta = \frac{\text{opp}}{\text{Adj}}$$

$$\tan \theta = \frac{30}{25} = 1.2$$

$$\theta = \tan^{-1} 1.2$$